

Lab 3 – 08.04.2026

During the class, we will follow a course that has been proposed by Google:

<https://developer.android.com/courses/android-basics-compose/course>

Tasks for the third lab:

1. Unit 3: Display Lists and Use Material Design

<https://developer.android.com/courses/android-basics-compose/unit-3>

2. Pathway 1: More Kotlin Fundamentals (*recommended, not mandatory*)

<https://developer.android.com/courses/pathways/android-basics-compose-unit-3-pathway-1>

Learn more Kotlin fundamentals such as generics, objects, extensions, and higher-order functions with collections.

a. Generics, Objects, and Extensions

<https://developer.android.com/codelabs/basic-android-kotlin-compose-generics>

b. Use Collections in Kotlin

<https://developer.android.com/codelabs/basic-android-kotlin-compose-collections>

c. Higher-Order Functions with Collections

<https://developer.android.com/codelabs/basic-android-kotlin-compose-higher-order-functions>

d. Practice: Generics and Collections

<https://developer.android.com/codelabs/basic-android-kotlin-compose-practice-classes-and-collections>

3. Pathway 2: Build a Scrollable List

<https://developer.android.com/courses/pathways/android-basics-compose-unit-3-pathway-2>

Learn how to display a list of data in an Android app and make it scrollable using Jetpack Compose.

a. Add a scrollable list

<https://developer.android.com/codelabs/basic-android-kotlin-compose-training-add-scrollable-list>

b. Change the app icon

<https://developer.android.com/codelabs/basic-android-kotlin-compose-training-change-app-icon>

- c. Practice: Build a grid (**mandatory**)
<https://developer.android.com/codelabs/basic-android-kotlin-compose-practice-grid>

4. Pathway 3: Build Beautiful Apps

<https://developer.android.com/courses/pathways/android-basics-compose-unit-3-pathway-3>

Learn how to apply Material Design, add simple animations, and test for accessibility in your Android apps.

- a. Material Theming with Jetpack Compose
<https://developer.android.com/codelabs/basic-android-kotlin-compose-material-theming>
- b. Simple Animation with Jetpack Compose
<https://developer.android.com/codelabs/basic-android-kotlin-compose-woof-animation>
- c. Testing for Accessibility
<https://developer.android.com/codelabs/basic-android-kotlin-compose-test-accessibility>
- d. Practice: Build Superheroes app
<https://developer.android.com/codelabs/basic-android-kotlin-compose-practice-superheroes>
- e. Project: Create a 30 Days App (**mandatory**)
<https://developer.android.com/codelabs/basic-android-kotlin-compose-30-days>

As a result of the work please take the Quiz at the end of each Pathway.

Submission requirements:

- Use the **private** repository named MAD created during Lab 2.
- Inside the repository maintain the following folder structure:

```
MAD/  
  UNIT3/  
    Grid                (Practice: Build a grid)  
    30DaysApp/          (Project: Create a 30 Days App)
```

- The instructor should already be a collaborator (rmaciaszczyk@zut.edu.pl). If not, add them using that email address.
- Submit the repository link as a response to this assignment in **Microsoft Teams**.